Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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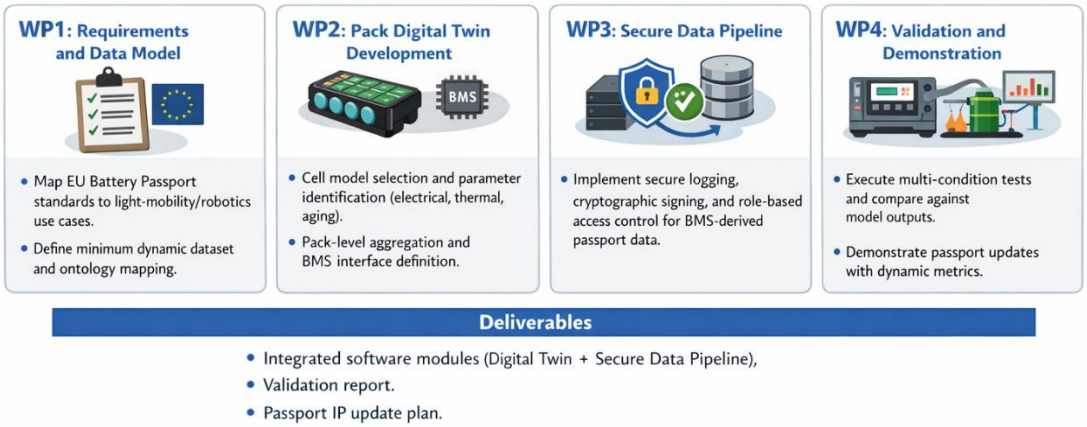
2. Technology Overview

Title of Proposed Technology	Digital Battery Passport-ready BMS and Pack Digital Twin for Light-Mobility and Robotics Li-ion Battery Packs			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Bio/Health ( ), Hydrogen ( ), Future Mobility ( ), Cyber Security ( ), AI (√), Next Generation Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics)		Information & Communications	
	Medium Category	Small Category	Medium Category	Medium Category
	Batteries	Secondary Batteries	Information Security	Common Security Technologies
Concept of Proposed Technology	Electrified mobile platforms such as service robots, logistics robots, drones, and micro-mobility vehicles increasingly rely on high-mix battery packs that must operate safely, efficiently, and transparently across diverse duty cycles and environments. At the same time, emerging EU regulatory frameworks, including the Battery Regulation and Battery Passport requirements, demand standardized, auditable metrics describing battery performance, health, and lifecycle events throughout use.			

	  This project addresses these technical and regulatory challenges by developing an integrated battery management and digital modelling solution. The concept combines cyber-secure BMS firmware and high-fidelity data logging with a validated electro-thermal and degradation pack digital twin capable of generating regulated, dynamic battery metrics. These outputs include energy efficiency indicators, state-of-health proxies, and structured event histories suitable for Battery Passport reporting. The approach explicitly integrates physics-based modelling, data-driven estimation, and controlled validation testing under realistic environmental and operational conditions to ensure traceability, robustness, and regulatory relevance [1][2][3]
Rationale for Joint R&D Project	The Korean SME partner designs and manufactures customized battery packs and BMS for diverse applications. To scale into Europe, future products must meet increasingly strict safety, cybersecurity and data-traceability requirements and provide trustworthy in-use performance metrics. TNO contributes validated models, test infrastructure and data-space compliant architectures to make the SME's BMS outputs auditable, comparable and portable across OEM and service ecosystems. [1][2]
Key Words	1. Battery passport 2. BMS data integrity 3. Pack digital twin 4. Degradation modelling 5. Validation testing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	1) Define a Battery Passport data model and minimum dynamic dataset for light-mobility and robotics packs, including negative event logging and energy-efficiency metrics. 2) Develop and validate a computationally efficient electro-thermal-degradation model (pack digital twin) that runs in near real time and interfaces with BMS data. 3) Implement secure data capture and tamper-evident logging from BMS to passport interface. 4) Demonstrate on representative pack prototypes and duty cycles provided by the Korean SME partner.			
R&D Target Specifications	Model accuracy targets (to be finalized in Phase 1): • Temperature prediction error: ≤2°C (cell surface) under representative dynamic load profiles. • Voltage prediction error: ≤50 mV per cell under representative profiles. • SOH proxy (capacity/energy) estimation error: ≤5% versus reference lab tests. • Event detection: detect and time-stamp over-temperature, over-current, over/under-voltage, and shock events as available. All targets are to be verified against controlled tests in TNO facilities and SME validation benches.			
R&D Plan	 **Work Package 1: Requirements and data model (Month 0 - 6)** • Map EU Battery Passport and related standards requirements to light-mobility/robotics pack use cases. • Define minimum dynamic dataset and ontology mapping. **Work Package 2: Pack digital twin development (Month 4 - 24)** • Cell model selection and parameter identification (electrical, thermal, aging). • Pack-level aggregation and BMS interface definition. **Work Package 3: Secure data pipeline (Month 16 - 28)**			

	• Implement secure logging, cryptographic signing and role-based access control for BMS-derived passport data. **Work Package 4: Validation and demonstration (Month 28 - 36)** • Execute multi-condition tests (temperature, load, altitude if relevant) and compare against model outputs. • Demonstrate passport record updates with dynamic metrics during in-use simulated cycles. Timeline and milestones will be detailed after Phase-1 scoping with the SME and TIPA process. **Validation infrastructure (TNO):** • Multichannel cell testing: 36 independent channels, 0-6 V and ±100 A (parallelisable), environmental range -40°C to +180°C and 10-98% RH. • Multichannel EIS: 8 channels, -10 V to +10 V and ±16 A (AC/DC), up to 100 kHz (above 5 A up to 10 kHz). • Pack testing: programmable bi-directional battery simulator 30-750 VDC, ±400 A, 300 kW; drive-in climatic chamber -60°C to +150°C with 5-95% RH; climatic-altitude chamber -45°C to +55°C and 628 mbar to 1 bar (up to 4000 m altitude) with wind generator 120 km/h. These facility specifications are taken from the TNO Powertrains introduction deck. [4]

4. R&R (Roles and Responsibilities),

Expected R&R of Research Institute	TNO is expected to: • Translate EU Battery Regulation and Battery Passport requirements into technical and data-level specifications applicable to light-mobility and robotics battery systems. • Define the minimum dynamic dataset required for Battery Passport compliance, including performance, health, and event-related metrics. • Develop physics-based and hybrid electro-thermal–degradation models forming the core of the pack-level digital twin. • Implement methodologies for real-time or near-real-time estimation of battery states (temperature, SOH proxies, efficiency indicators). • Design the digital architecture for secure data handling, including traceability, integrity, and controlled data access aligned with European data space principles. • Develop interfaces between BMS data streams and digital twin models. • Perform laboratory validation using controlled electrical, thermal, and environmental test facilities. • Verify model accuracy against experimental results and define acceptance thresholds. • Provide technical documentation, modeling guidelines, and validation reports. Support regulatory alignment and future certification preparation related to Battery Passport and data traceability requirements.
Expected R&R of Korean SME Partner	Korean SME partner is expected to: • Provide representative battery pack designs, system specifications, and application requirements. • Supply BMS hardware details, communication protocols, and firmware access required for integration. • Implement or adapt BMS firmware to enable required data logging and event recording functions. • Manufacture prototype battery packs for validation and demonstration activities. • Support system integration of digital twin interfaces and data export mechanisms. • Conduct complementary testing using in-house facilities and operational profiles. • Provide feedback on manufacturability, cost constraints, and practical deployment considerations. • Contribute operational data or usage scenarios for validation of models and algorithms. • Lead commercialization planning and preparation for EU market deployment. • Apply project results to future products targeting Battery Passport compliance and export readiness.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• A validated pack digital twin and associated parameterisation workflow for high-mix Li-ion battery packs. • A Battery Passport-ready data interface specification and reference implementation for BMS event and performance data. • Demonstrated secure, auditable in-use metric generation (efficiency, SOH proxies, event history) suitable for passport updates. • Reduced engineering iteration time for new pack variants through simulation-led design and test planning. • Improved safety and compliance readiness (including cybersecurity and traceability) for EU market access.

6. References

(1) TNO. "Batterij Lab" (Battery Lab facility page). Accessed 18 Dec 2025. https://www.tno.nl/nl/technologie-wetenschap/faciliteiten/batterij-lab/

(2) Rath, S.; Medina, R.; Wilkins, S. "Real-time optimal charging strategy for a fleet of electric vehicles minimizing battery degradation." TNO publication (PDF), 2023. Accessed 18 Dec 2025. https://publications.tno.nl/publication/34641555/EemswK/rath-2023-real.pdf

(3) Wilkins, S. "Enhanced battery model including temperature effects." EVS27 International Battery, Hybrid and Fuel Cell Electric Vehicle Symposium, Barcelona, Spain, 17-20 Nov 2013. TNO Repository record. Accessed 18 Dec 2025. https://repository.tno.nl/SingleDoc?docId=39157

(4) TNO Powertrains. "TNO Powertrains introduction" (battery lab and climatic chamber specifications). Presentation deck. Accessed 18 Dec 2025. (Internal file)

(5) Battery Pass Consortium. Battery Passport Content Guidance (Q&A / Content Guidance). 2024. https://thebatterypass.eu/wp-content/uploads/q-a_content-guidance.pdf (accessed 2025-12-17).

(6) TNO. International Data Spaces (IDS). https://www.tno.nl/en/technology-science/technologies/international-data-spaces/ (accessed 2025-12-17).

(7) TNO. TNO Security Gateway (TSG) Documentation: Overview. https://tno-tsg.gitlab.io/docs/overview (accessed 2025-12-17).

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization